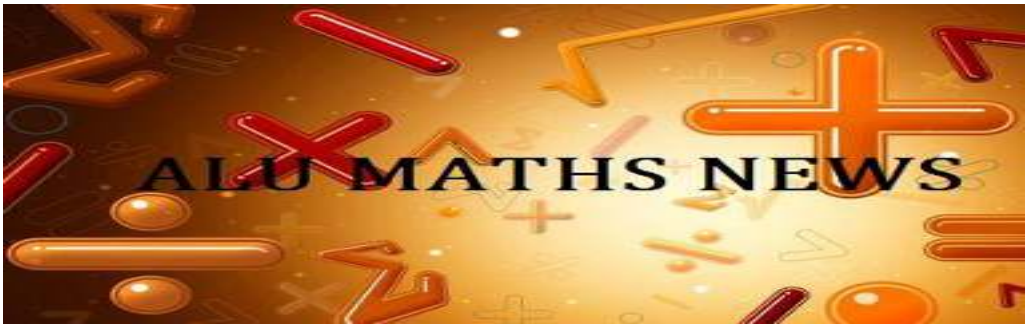

ALU SCHOOL OF MATHEMATICS

Volume 1, Issue 8 **News Letter** *Since: 09/12/2016*

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We are delighted to bring to you this issue of ALU Mathematics News, a monthly newsletter dedicated to the emerging field of Mathematics. This is the first visible —output from the Department of Mathematics, Alagappa University. We are committed to make ALU Mathematics News a continuing and effective vehicle to promote communication, education and networking, as well as stimulate sharing of research, innovations and technological developments in the field. However, we would appreciate your feedback regarding how we could improve this publication and enhance its value to the community. We are keen that this publication eventually grows beyond being a mere —news letter to become an invaluable information resource for the entire Mathematics community, and look forward to your inputs to assist us in this endeavor.

Dr. N. Anbazhagan

MATHEMATICS

Mathematics is the study of topics such as quantity, structure, space, and change. There is a range of views among mathematicians and philosophers as to the exact scope and definition of mathematics. When mathematical structures are good models of real phenomena, then mathematical reasoning can provide insight or predictions about nature. Through the use of abstraction and logic, mathematics developed from counting, calculation, measurement, and the systematic study of the shapes and motions of physical objects.

SYMBOLS MAKES PICTURE

tho;f;if fzpjk;

tho;f;if xU fzpjk; ...!

,jpy; Njitfis $l;lpdhy;
Kaw;rpfs; jhNd ngUFk; !
Kaw;rpfis $l;lpdhy;
mDgtq;fs; jhNd ngUFk; !

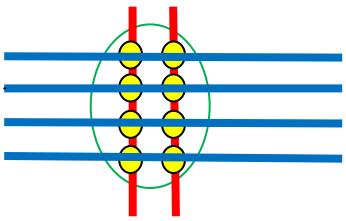
njspthd ghijia tFj;J tpl;lhy;
Ntz;lhjit jhNd fopfpd;wd !

fw;f Ntz;lpait kl;Lk;
tho;f;ifapy; mjpfk;

gpwh; Kaw;rpia fld; thq;fp
jdJ cwf;fj;ij fopg;gJk;
rKjha gugug;Gfis rhjfkhf;Fk;
jrk];jhdq;fSk; fw;W tpl;lhy;
tho;f;if fzpjj;jpw;F
tpil jhNd fpilj;J tpLk; !

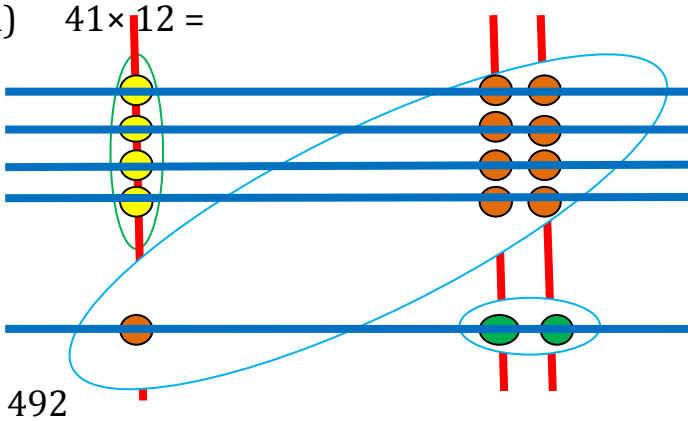
EASY MULTIPLICATION

(i) 4 × 2 =



Ans : 8

(ii) 41× 12 =



Ans : 492

SYMBOLS MAKES TREE

Foe;ij [ddkhFk; khjq;fs;

gj;jhf ,Uf;fl;Lk;

Nfhtpypy; fpufq;fs;

Xd;gjhf ,Uf;fl;Lk;

tuk; jUk; yl;Rkpfs;

vl;lhf ,Uf;fl;Lk;

thuj;jpy; ehl;fs;

Vohf ,Uf;fl;Lk;

mUs;KUfd; giltPL

Mwhf ,Uf;fl;Lk;

mfpyj;jpy; G+jq;fs;

Ie;jhf ,Uf;fl;Lk;

rJuj;jpd; gf;fq;fs;

ehd;fhf ,Uf;fl;Lk;

rq;fj; jkpo;fs;

%d;whf ,Uf;fl;Lk;

G+kpf;Fj; JUtq;fs;

,uz;lhf ,Uf;fl;Lk;

kf;fspd; vz;zq;fs;

xd;whf ,Uf;fl;Lk;

NghH kl;Lk; ,t;Tyfpy;

G+[;[pakha; ,Uf;fl;Lk;

nghq;Ffpw md;G kl;Lk;

vz;zpw;W ,Uf;fl;Lk;.

THE GREATEST THINGS

37037 × 3 = 111111
37037 × 6 = 222222
37037 × 9 = 333333
37037 × 12 = 444444
37037 × 15 = 555555
37037 × 18 = 666666
37037 × 21 = 777777
37037 × 24 = 888888
37037 × 27 = 999999

WONDER IN MATHS

12345679 × 9 = 111111111
12345679 × 18 = 222222222
12345679 × 27 = 333333333
12345679 × 36 = 444444444
12345679 × 45 = 555555555
12345679 × 54 = 666666666
12345679 × 63 = 777777777
12345679 × 72 = 888888888
12345679 × 81 = 999999999

kdf; fzf;F

fhy;FNyl;liuAk; Njlf; $lhJ. fhfpjj;jpYk; nra;J ghh;f;ff; $lhJ.
kdjpw;Fs; $l;L Kaw;rp nra;aTk;.

1. cq;fsplk; 1000 &gha; cs;sJ.
2. mjDld; 40 &gha; $l;lTk;.
3. ,d;ndhU 1000 &gha; $l;lTk;.
4. mjDld; 30 &gha; $l;lTk;.
5. kPz;Lk; xU 1000 &gha; $l;lTk;.
6. mjDld; 20 &gha; $l;lTk;.
7. kPz;Lk; xU 1000 &gha; $l;lTk;.

filrpahf xU 10 &gha; $l;;lpdhy; vd;d tUk;?

Ans: 5000 te;jhy; jtW. rupahd tpil 4100.

FUNNY MATHS

24-1=24....!?

24 pieces total chocolate :

Cut the chocolate as shown below:

Divide the chocolate as shown below:

Replace the chocolate as shown below:

Now join the separate pieces, you will have 24 pieces and an extra one piece :

FUN IN MATHS

(i) $37 \times 3 = 111$
$37 \times 6 = 222$
$37 \times 9 = 333$
$37 \times 12 = 444$
$37 \times 15 = 555$
$37 \times 18 = 666$
$37 \times 21 = 777$
$37 \times 24 = 888$
$37 \times 27 = 999$

(ii) $15873 \times 7 = 111111$
$15873 \times 14 = 222222$
$15873 \times 21 = 333333$
$15873 \times 28 = 444444$
$15873 \times 35 = 555555$
$15873 \times 42 = 666666$
$15873 \times 49 = 777777$
$15873 \times 56 = 888888$
$15873 \times 63 = 999999$
$15873 \times 70 = 1111110$

(iii) $123456789 \times 9 \times 1 = 111111111$
$123456789 \times 9 \times 2 = 222222222$

123456789 × 9 ×3 = 333333333
123456789 × 9 × 4 = 444444444
123456789 × 9 × 5 = 555555555

WHO AM I

If it is a 9 letter word 123456789. Solve if you can ?
It is a 9 letter word 123456789
If you lose it you die,
If you have 234, you can 1234
56 is one type of disease , 89 indicates exact location and time, 2 &7 are same letter and 3 & 8 are same letters & 3 & 8 are same letters. 5 & 9 are same letters.

Guess the word ? ? ?

It's a challenge for all masterminds !
Ans :

HEART BEAT.

FUNNY MULTIPLICATION

123456789 × 9 = 1111111101
123456789 × 18 = 2222222202
123456789 × 27 = 3333333303
123456789 × 36 = 4444444404
123456789 × 45 = 5555555505
123456789 × 54 = 6666666606
123456789 × 63 = 7777777707
123456789 × 72 = 8888888808
123456789 × 81 = 9999999909

CALENDER

What was the day of the week on 16th July, 1776 ?

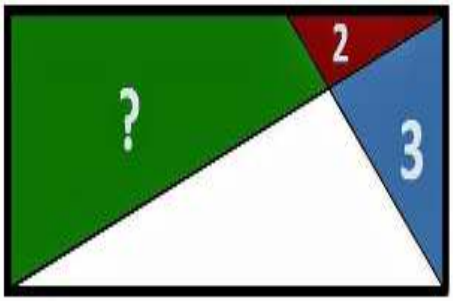
Solution 16th July, 1776 = (1775 years + period from 1.1.1776 to 1.7.1776) Number of odd days in 1600 Years= 0Number of odd days in 100 Years= 5 75 Years = 18 leap Years + 57 ordinary Years = (18 ×2 + 57×1) odd days = 93 odd days1775 Years have = (0+5+2) odd days = 7 odd days = 0 odd days January, February, March, April, May, Jun, July(31 + 29 +31+ 30 +31+30 +16) = 198 days

198 days = (28 weeks + 2 days) = 2 odd days
Total numbers of odd days = (0 + 2) = 2
Hence the required day is Tuesday.
Ans : Tuesday

PUZZLES WITH IMAGES

1) There is a figure below (a rectangle). You can see different colors depicting different regions of the figure. The labels on the top of a region defines the area of that region.

Can you find the area of the green shaded region labelled with a question mark?



Solution

This problem can be solved with similar triangles property.
The White triangle is similar to the Red oneNow Ratio of sides is 2:3 (as ratio of areas = ratio of based if the height is same)Seeking that, White triangle = 9/4 * Yellow triangle = 9/2
Following all the above, the Green region = 5.5 units

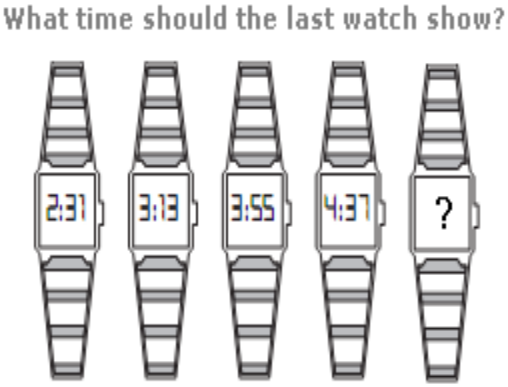
2) Can you find the correct number which should replace the question mark in the picture below

Solution- 3

3) Can you solve the maths puzzle by replace the question mark with the correct number ?

Solution- 8

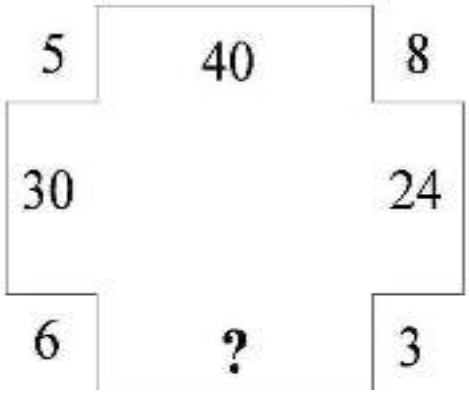
4) Based on previous watches, what time should the last watch show ?



Solution- 5:19

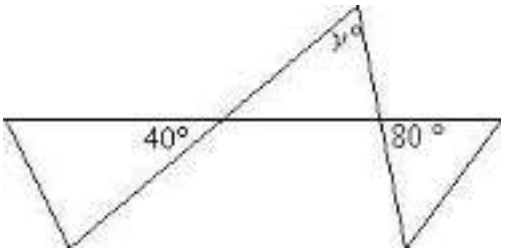
Starting with the watch on the left, add 42 minutes to the time shown to give the time on the next watch to the right

5) Can you replace the "?" mark with the correct number ?



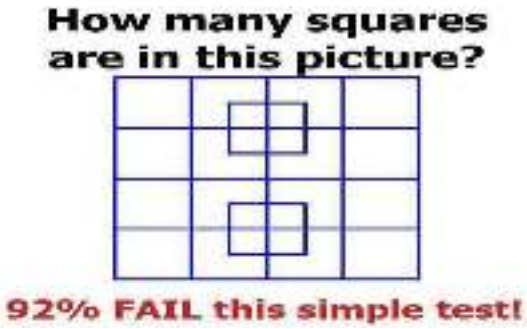
Solution- 18

6) In the picture below, can you identify the value of angle 'Y'



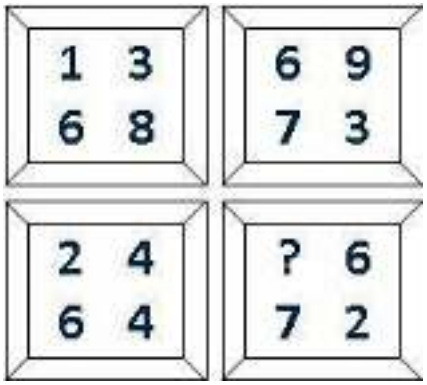
Solution- 60

7) How many squares are there in the following figure ?



Solution- 40

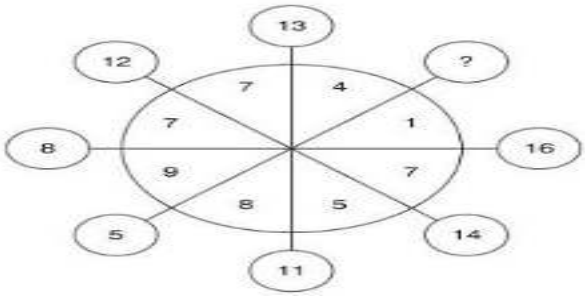
8) Which number should replace the question mark in the picture below ?



Solution: 4

6*3 equals 18
7*9 equals 63
6*4 equals 24
7*6 equals 42, ? should be replaced by 4.

9) Which number should replace the question mark In The Circle Below ?



Solution- 17

It is the sum of the two digits(9 + 8) in opposite quadrant.

10) A 12-year footballer Lavy was offered a contract with a club name "Flyball" that if he will practice 1 hour extra after school he will be offered 11$ after every week.Lavy, however, suggested an alternative paid method in which He will be paid just a penny on his first day. Two pence will be paid on the second day, Four pence will be paid on the third day.And so on till 11th day. Should the club accept the offer of Lavy?

Solution

Flyball should pay Lavy $11 instead of accepting his counter offer as in that case the club will end up paying

almost double the amount.

Day 1	01
Day 2	02
Day 3	04
Day 4	08
Day 5	16
Day 6	32
Day 7	64
Day 8	128
Day 9	256
Day 10	512
Day 11	1024

♠♠♠♠♠♠